

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Artificial Intelligence & Search Algorithms | Course Code: 21AI52 | Credits: 3

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** Trace the A* Search algorithm on a given navigation graph from Start node S to Goal node G with step-by-step OPEN and CLOSED lists, verifying that heuristic $h(n)$ is admissible.

[8 Marks]